



Afonso Morgado Mota

21/03/2000

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Skills

Adaptability

Fast Learner

Effective Time Management

Communication Skills

Responsibility & Reliability

Proficiency in productivity tools

Languages

Portuguese

English

Italian

Spanish

Programming Languages

Python

C#

ImageJ Macro Language

R

ShellScript

Pascal

SQL

Profile

Highly motivated early-career researcher passionate about Astrobiology. Experienced in academic research projects, as well as in presenting them in acclaimed national and international conferences and seminars. Studied and performed research in leading European institutions in Portugal, Germany and Italy. Excellent soft skills.

Education

PhD in Astronomy, Faculty of Sciences, University of Porto

September 2024 — Present

Master's in Biology of Extreme Environments, Astrobiology Curriculum, University of Naples "Federico II"

September 2021 — July 2023

Bachelor's in Biology, Faculty of Sciences, University of Lisbon

September 2018 — July 2021

Minor in Physics, Faculty of Sciences, University of Lisbon

September 2020 — July 2021

Research Experience and Internships

Institute of Astrophysics and Space Sciences (IA), Porto, Portugal

September 2022 — Present

Researcher on exoplanet astrophysics and astrobiology.

Interdisciplinary Centre of Marine and Environmental Research (CIIMAR), Porto, Portugal

September 2024 — Present

Researcher on extreme environment microbiology and microbial ecology.

Aerospace Microbiology Group, Institute of Aerospace Medicine, German Aerospace Center (DLR), Cologne, Germany

September 2022 – July 2023

Developed of my master's thesis project in astrobiology under supervision of Dr. Marta Cortesão, Dr. Nuno Santos and Dr. Marco Moracci.

Giovannelli Lab, University of Naples “Federico II”, Italy

March 2022 – May 2022

Received training and assisted the development of a Microbiology project with the laboratory’s research group.

Marco Moracci’s Lab, University of Naples “Federico II”, Italy

November 2021 — December 2021

Received training and aided in biotechnological projects studying extremophilic enzymes.

Marine and Environmental Sciences Center (MARE), FCUL, Portugal

October 2019 — March 2020

Laboratory work for ecological studies of Portuguese aquatic environments.

"Robótica no Espaço" (Robotics in Space) – Laboratory for Robotics and Engineering Systems (LARSyS), IST-ID, Instituto Superior Técnico, Lisbon, Portugal

July 2017

Internship where we learned how to program controls for the movement of a simulated satellite. My team and I won the prize for the 2nd best performing program.

Group Of Lasers and Plasmas (GOLP), Instituto Superior Técnico, Lisbon, Portugal

June 2016

Internship for learning about physics research.

Conference Attendance

Characterizing Habitable Exoplanets With Interdisciplinary Expertise (CHEWIE) Workshop, Lorentz Center, Leiden, The Netherlands

28th April – 1st May 2026

Rocky Worlds 4 Conference, Energy Academy Europe, Groningen, The Netherlands

19th-23rd January 2026

XVII Portuguese Conference of Polar Sciences, Lusófona University, Lisbon, Portugal

14th-15th November 2025

EANA 2025 Conference, Instituto Superior Técnico, Lisbon, Portugal

21st-24th October 2025

Blue Think Conference 2025, CIIMAR, Porto, Portugal

10th September 2025

“Astrobiology: Between Stars and Microorganisms”, Center of Astrophysics of the University of Porto, Porto, Portugal

24th-25th May 2025

XVI Portuguese Conference of Polar Sciences, Almeida, Portugal

24th-25th October 2024

EANA 2023 Conference, University Carlos III of Madrid, Spain

19th-22nd September 2023

Launching Space Biology (LSB) Workshop, University of Colorado Boulder, Boulder, USA

22nd-26th May 2023

Publications

- **Mota, A.**, Schiele, A., de Sousa, I. S., & Cortesão, M. (2026). Open-source digital tools for filamentous fungi analysis. *Journal of Microbiological Methods*, 107465. <https://doi.org/10.1016/j.mimet.2026.107465>
- Schiele, A., Timofeev, S. M., **Mota, A.**, & Cortesão, M. (2025). Fungus *Penicillium rubens* isolated from the International Space Station (ISS) shows faster colony growth and better spore resistance to cosmic radiation than type strain. *International Journal of Astrobiology*, 24, e13. <https://doi.org/10.1017/S147355042510013X>
- **Mota, A.**, Koch, S., Matthiae, D., Santos, N., & Cortesão, M. (2025). How habitable are M dwarf exoplanets? Modeling surface conditions and exploring the role of melanins in the survival of *Aspergillus niger* spores under exoplanet-like radiation. *Astrobiology*, 25(3), 161–176. <https://doi.org/10.1089/ast.2024.0023>
- Marques, C. S., **Mota, A.**, Belvedere, ... & Dufourq, E. (2025). Deep tracks: Using deep learning and procedurally simulated data for automated vertebrate footprints classification. *Ecological Informatics*, 103523. <https://doi.org/10.1016/j.ecoinf.2025.103523>
- Koch, S. M.; Pohl, C.; Siontas, O.; Cortesão, M; **Mota, A.**; Runzheimer, K.; Jung, S.; Rebrošová, K.; Siler, M.; Moeller, R., & Meyer, V. (2023). *Aspergillus niger* as a cell factory for the production of pyomelanin, a molecule with UV-C radiation shielding activity. *Frontiers in Microbiology*, 14. <https://doi.org/10.3389/fmicb.2023.1233740>

Communication and Conference Talks

- **Organizing conference (SOC and LOC)** Blue Think Conference 2026, Algarve, Portugal, September 2026
- **Upcoming talk** at the **Exoplanets 6** conference, Porto, Portugal, June 2026.
- **Organizing conference (LOC)** Exoplanets 6, Porto, Portugal, June-July 2026.
- **Talk** at the **XVII Portuguese Conference of Polar Sciences** in Lisbon, Portugal, November 2025
- **Organized conference (SOC and LOC)** “Astrobiology: Between Stars and Microorganisms” at the Center of Astrophysics of the University of Porto, May 2025.
- **Class** to the students of the course “**Astrobiology**”, Master’s in Astronomy and Astrophysics, University of Porto, Portugal
- **Talk** at the **XVI Portuguese Conference of Polar Sciences** in Almeida, Portugal, October 2024
- Creator of **AstroBioLusitanos**, a group joining Portuguese Astrobiology graduates. We regularly engage in science communication on our Instagram page (@astrobiopt). I have given **>10 talks and workshops in schools and universities** with this initiative.

- **Yearly class** to the students of the course “**Emergence of life and exobiology**”, Master’s in Biology of Extreme Environments (curriculum Astrobiology), University of Naples “Federico II”. Online. 2023, 2024, 2025.
- **Seminar** to the **Institute of Astrophysics and Space Sciences (IA)** at the Astrophysics Center of the University of Porto (CAUP). 22-11-2023.
- **Class** to the students of the course “**Environmental Microbiology**” from the Faculty of Sciences of the University of Porto. 26-10-2023. Online.
- **Talk** at the “Life Sciences and Extremophiles” Session of the **European Astrobiology Network Association (EANA) 2023 Conference**. 22-09-2023.
- **Talk** at the “Space Factor” Session of the **European Astrobiology Network Association (EANA) 2023 Conference**. 20-09-2023.

Highlighted Awards and Distinctions

Final selection round for SOLIS100 study (top 1.5%) – DLR/ESA

February 2026

Selection process and standards similar to the ones applied for choosing ESA astronauts.

2nd most voted poster – Rocky Worlds 4

January 2026

Poster: Cataloguing the Limits of Life and Modelling Potential Niches on Mars and Exoplanets

Poster Award – EANA 2025

October 2025

Poster: Cataloguing the Limits of Life and Modelling Potential Niches on Mars and Exoplanets

3rd prize in the Space Factor contest – EANA 2023

September 2023

Presentation: Beyond the Solar System: Search for Microbial Life on Exoplanets and the Role of Melanin

Best Poster – VAAM Microbiome Symposium 2022

December 2022

Poster: Life beyond our backyard: exploring microbial habitability on exoplanets

Represented Portugal in International Science and Engineering Fair (ISEF), Pittsburgh, USA

May 2018

Project: Assessment of the levels of mercury in students aged 12 to 18.

Other Positions

Large Interferometer for Exoplanets (LIFE) Science Team member

May 2026 – Present

Involved in research projects directly related to the development of the future flagship telescope LIFE.

Secretary and Executive Committee Member of Astrobiology Graduates of Europe (AbGradE)

January 2026 – Present

Main organization joining and representing early career researchers in Astrobiology in Europe.